
📖 Chapter 11 – 2D Transforms in CSS: Ghumao, Hilao, Badhao! 🌀

❑ What is CSS Transform?

`transform`: ek CSS property hai jo **element ko modify/move** karti hai — without changing its actual position in HTML.

Matlab element wahi rehta hai, lekin **rotate, zoom, move, tilt** sab ho sakta hai 🤖

2D me sirf X aur Y axis ka use hota hai.

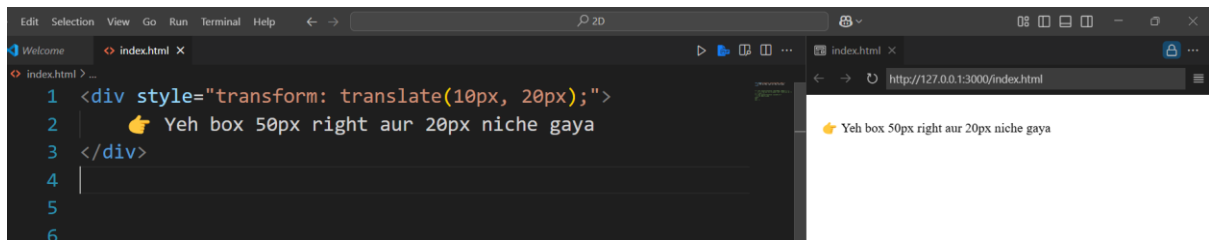
✨ 2D Transform Properties:

Property	Kaam kya karta hai
<code>translate(x, y)</code>	Element ko move karta hai
<code>rotate(deg)</code>	Element ko ghumata hai (deg me)
<code>scale(x, y)</code>	Size badhata/ghataata hai
<code>skew(x, y)</code>	Element ko tilt karta hai (angle se)
<code>matrix()</code>	Sab kuch ek hi function me mix karta hai (advanced)

🔗 1. Translate – Move Karna

```
<div style="transform: translate(50px, 20px);">  
  ☞ Yeh box 50px right aur 20px niche gaya  
</div>
```

❑ Positive value → right/bottom
Negative value → left/top



🌀 2. Rotate – Ghuma Dena

```
<div style="transform: rotate(45deg);">
  🖱️ Yeh box 45° ghoom gaya
</div>
```

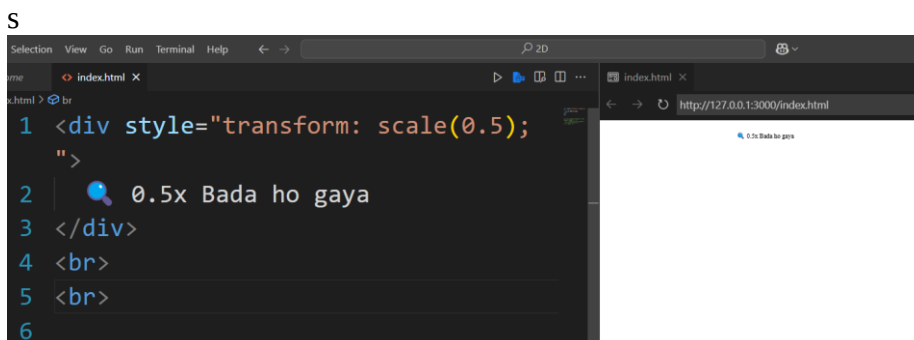
□ -45deg doge to ulta ghoomega (anti-clockwise)



🔍 3. Scale – Zoom In/Out

```
<div style="transform: scale(1.5);">
  🔍 1.5x Bada ho gaya
</div>
```

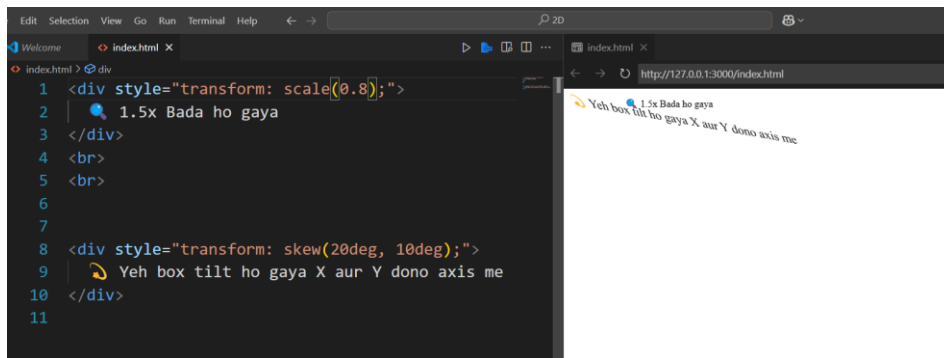
□ scale(2) = 2x bada
scale(0.5) = 50% chhota



🔗 4. Skew – Tilt Karna

```
<div style="transform: skew(20deg, 10deg);">
```

🔗 Yeh box tilt ho gaya X aur Y dono axis me
</div>

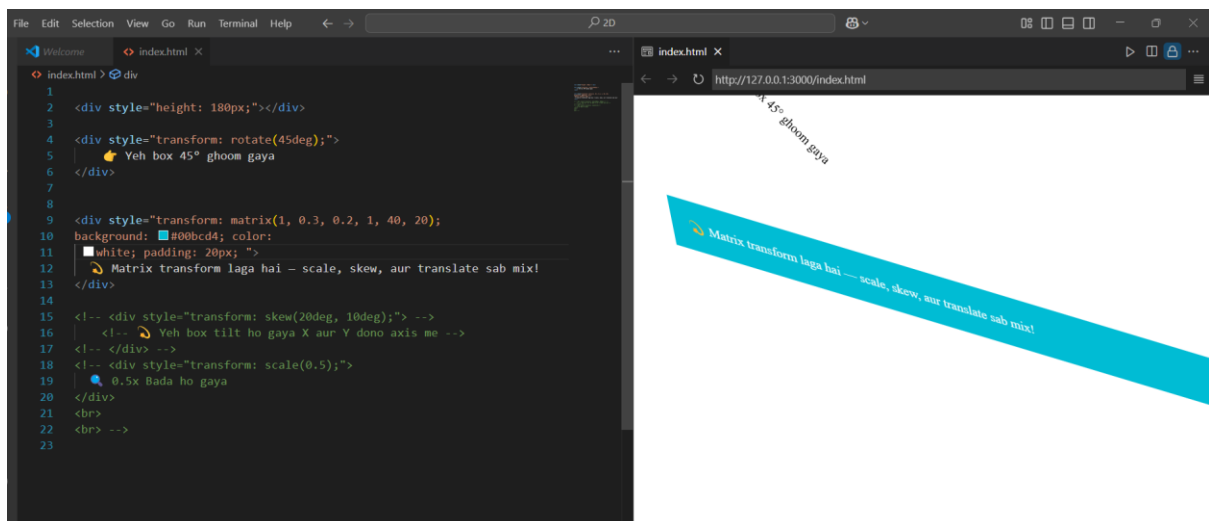


🔗 5. Matrix – Advance Combo

Rarely used – mix of all above in ek hi line.

```
transform: matrix(scaleX, skewY, skewX, scaleY, translateX, translateY);
```

🔗 Advanced devs ke liye hai – abhi ke liye chhodo 😊



🏠 Example with Hover Effect:

```
<style>
.hover-box {
  width: 100px;
  height: 100px;
  background: tomato;
  transition: 0.3s;
}

.hover-box:hover {
  transform: rotate(15deg) scale(1.2);
}
</style>
```

```
<div class="hover-box">🌀 Hover Me</div>
```

☐ Hover pe zoom bhi hoga, ghoomega bhi – mast stylish!

📌 Notes:

- `transform`: CSS me **inline bhi likh sakte ho** ya class me bhi
 - Combine bhi kar sakte ho → `transform: rotate(45deg) scale(1.5);`
 - Use `transition`: to make it **smooth**
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📱 Summary:

- `translate(x, y)` → Move karo
 - `rotate(deg)` → Ghumao
 - `scale(x, y)` → Zoom in/out
 - `skew(x, y)` → Tilt karo
 - Combine bhi kar sakte ho ek line me
 - Use `transition`: for animation effect
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